

William Jin

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Education

University of Waterloo | Bachelor of Software Engineering, Honours *Waterloo, ON | Sep 2024 – May 2029*

- Certifications: Microsoft Azure Fundamentals (AZ-900) and Azure AI Fundamentals (AI-900)
- Relevant Coursework: Data Structures, Software Design (ADTs), Databases, Digital Computers, Statistics

Experience

Software Engineer Intern | Correctional Service of Canada *Ottawa, ON (Remote) | Jan 2026 – May 2026*

- Built a chatbot on Azure OpenAI and AI Search, letting **17,000+** officers and staff query **220+** policy directives
- Raised search accuracy **15%** and answer grounding **10%** on 300 prompts by re-parsing PDFs to HTML on Azure
- Added a TypeScript citation panel linking each answer to its source, plus a rating flow logging officer feedback

Software Engineer Intern | UAlberta Energy Mechatronics Lab *Edmonton, AB | Jul 2025 – Sep 2025*

- Cut **200+ hours** of manual author lookups by automating citation tracking for 7,000+ papers with a Selenium scraper
- Mapped the results as a Folium choropleth shading regions by citation density, used in lab research talks
- Consolidated **5,000+** fuel cell experiment files into a SQL Server database, replacing manual folder search

Software Engineer Intern | Jindon International Ltd. *Waterloo, ON | Jun 2025 – Jul 2025*

- Built case search and intake forms in Java for a government benefits platform serving **~1M** users
- Optimized Oracle SQL query performance **10x** by eliminating redundant joins and adding targeted indexes
- Wrote JMockit unit tests across the platform's case-management modules and fixed defects in its UI

Projects

Tract | Personal Project *Feb 2026 – Mar 2026*

- Architected a git-style version-control engine for LLM context in Python: branching, three-way merge, and rebase
- Wrapped the engine in an async agent loop exposing **28** LLM tools, with token budgeting and auto-compression
- Persisted the object store in SQLite via SQLAlchemy, backed by a property-based and end-to-end pytest suite

Stitch | UofTHacks 13 *Jan 2026*

Winner: Best Use of Gemini API

- Built an agentic video editor in Next.js where a Gemini agent drives a live timeline through **14** custom editing tools
- Implemented natural-language clip retrieval with Twelve Labs, resolving each query to exact in-clip timestamps
- Designed a React 19 timeline where agent and manual edits share a single undo/redo history
- Backed the editor with Supabase Postgres and storage buckets, exporting cuts through FFmpeg in Next.js API routes

Solana Swift SDK *Oct 2025 – Nov 2025*

\$40K USD Solana Foundation grant

- Developed Solana's first official native Swift SDK, a typed iOS client for multi-wallet connection and signing
- Secured each wallet session with an X25519 deeplink handshake in TweetNaCl, so the app never sees a wallet key
- Layered a GraphRAG documentation agent over the SDK, indexing **600+** Swift symbols into a FalkorDB code graph

SolShare | Hack the North 2025 *Sep 2025*

1st Place: Cohere API Best Use; Solana Best Consumer Payment Experience

- Shipped a bill-splitting iOS app in Swift that turns a receipt photo into per-person totals
- Parsed receipts into line items with Cohere vision models in a self-critic loop that re-checks each extraction
- Built the backend on Firebase Cloud Functions and Firestore, settling each share on-chain as a Solana transfer

Skills

AI/ML: PyTorch, Hugging Face, Unsloth, LangChain, LangSmith, Agno, NumPy, Pandas

Languages: Python, TypeScript, JavaScript, Java, Swift, C++, SQL

Full Stack: React, Next.js, Tailwind CSS, SwiftUI, Node.js, Express.js, PostgreSQL, MySQL, Supabase

Cloud & Tools: Azure, GCP, Cloudflare, Vercel, Docker, OpenTelemetry, Git, Linux

Concepts: Backend Development, Full-Stack Development, Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Machine Learning, GenAI, RAG, AI Agents